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Homework 10
MATH 307

1. $\boxed{D = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}}$

AD will be the matrix that is the same as A with columns 1 & 4 swapped and columns 2 & 3 swapped.

$$AD = \begin{bmatrix} 2 & -1 & 5 & 0 \\ 0 & 6 & 1 & -4 \\ 3 & 3 & 1 & 1 \\ 0 & 8 & 2 & 9 \\ -4 & 3 & 2 & 5 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 5 & -1 & 2 \\ -4 & 1 & 6 & 0 \\ 1 & 1 & 3 & 3 \\ 9 & 2 & 8 & 0 \\ 5 & 2 & 3 & -4 \end{bmatrix}$$

3.

$$E_2 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & l_{32} & 1 & 0 & 0 \\ 0 & l_{42} & 0 & 1 & 0 \\ 0 & l_{52} & 0 & 0 & 1 \end{bmatrix} = \boxed{\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & -3/4 & 1 & 0 & 0 \\ 0 & -4/3 & 0 & 1 & 0 \\ 0 & -1/6 & 0 & 0 & 1 \end{bmatrix}}$$

$$l_{32} = -\frac{a_{32}^{(1)}}{a_{22}^{(1)}} = -\frac{9}{2(6)} = -\frac{3}{4}, \quad l_{42} = -\frac{a_{42}^{(1)}}{a_{22}^{(1)}} = -\frac{8}{6} = -\frac{4}{3}, \quad l_{52} = -\frac{a_{52}^{(1)}}{a_{22}} = -\frac{1}{6}$$

$$A_2 = E_2 A_1 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & -3/4 & 1 & 0 & 0 \\ 0 & -4/3 & 0 & 1 & 0 \\ 0 & -1/6 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & -1 & 5 & 0 \\ 0 & 6 & 1 & -4 \\ 0 & 9/2 & -15/2 & 1 \\ 0 & 8 & 2 & 9 \\ 0 & 1 & 12 & 5 \end{bmatrix} = \boxed{\begin{bmatrix} 2 & -1 & 5 & 0 \\ 0 & 6 & 1 & -4 \\ 0 & 0 & -29/4 & 4 \\ 0 & 0 & 2/3 & 43/3 \\ 0 & 0 & 71/6 & 17/3 \end{bmatrix}}$$

5. $\boxed{\text{No,}}$ the matrix is not row reduced because the '1' in row 2 has a nonzero element above it.

$$\boxed{A_{RREF} = \begin{bmatrix} 1 & 0 & 6 \\ 0 & 1 & 3 \end{bmatrix}}$$

a) $\boxed{\text{rank} = 2}$ because only two columns are mutually linearly independent

- $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ are, but not together with $\begin{bmatrix} 6 \\ 3 \end{bmatrix}$
- Pf. $6\begin{bmatrix} 1 \\ 0 \end{bmatrix} + 3\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \end{bmatrix}$

b) $\boxed{\dim(N(A_{RREF})) = 1}$

- #columns − rank = 3 − 2

c) Basis of null space $= \begin{bmatrix} -B \\ I_{n-r} \end{bmatrix} = \boxed{\begin{bmatrix} -6 \\ -3 \\ 1 \end{bmatrix}}$ or $\left\{\begin{bmatrix} -6 \\ -3 \\ 1 \end{bmatrix}\right\}$

The matrix isn't square, therefore it is NOT invertible and its RREF isn't the identity matrix.

7. $A = LU$, let $Ly = b$:

$$\begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 13 \end{bmatrix}, \quad \underline{y_1 = 2}, \quad 3y_1 + y_2 = 13, \quad \underline{y_2 = 7}$$

If $Ly = b$ and $LUx = b$

$Ux = y$

$$\begin{bmatrix} 1 & -1 & 2 \\ 0 & -2 & -1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 7 \end{bmatrix},$$ the system is unsolvable: 2 equations & 3 variables

$x_1 - x_2 + 2x_3 = 2$, $-2x_2 - x_3 = 7$

$\underline{x_3 = -2\overset{\alpha}{x_2} - 7}$, $x_1 - x_2 - 4x_2 - 14 = 2$, $x_1 - 5x_2 = 16$

$\underline{x_1 = 5\overset{\alpha}{x_2} + 16}$, $\underline{\text{let } \alpha \text{ be a free variable, } \overset{x_2}{\alpha} \in \mathbb{R}}$

let $x_2 = \alpha$

$$\boxed{x = \begin{bmatrix} 5\alpha + 16 \\ \alpha \\ -2\alpha - 7 \end{bmatrix}}$$

8.

$$A = R^T R = \begin{bmatrix} r_{11} & 0 & 0 \\ r_{12} & r_{22} & 0 \\ r_{13} & r_{23} & r_{33} \end{bmatrix}\begin{bmatrix} r_{11} & r_{12} & r_{13} \\ 0 & r_{22} & r_{23} \\ 0 & 0 & r_{33} \end{bmatrix}$$

$r_{11} = \sqrt{a_{11}} = \underline{1}$, $r_{12} = \dfrac{a_{12}}{r_{11}} = \underline{2}$, $r_{13} = \dfrac{a_{13}}{r_{11}} = \underline{1}$

$r_{22} = \sqrt{a_{22} - r_{12}^2} = \sqrt{13 - 2^2} = \sqrt{9} = \underline{3}$, $r_{23} = \dfrac{a_{23} - r_{12}r_{13}}{r_{22}} = \dfrac{5-2}{3} = \underline{1}$

$r_{33} = \sqrt{a_{33} - r_{13}^2 - r_{23}^2} = \sqrt{18 - 1^2 - 1^2} = \sqrt{16} = \underline{4}$

$$\boxed{A = R^T R = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 1 & 1 & 4 \end{bmatrix}\begin{bmatrix} 1 & 2 & 1 \\ 0 & 3 & 1 \\ 0 & 0 & 4 \end{bmatrix}}$$